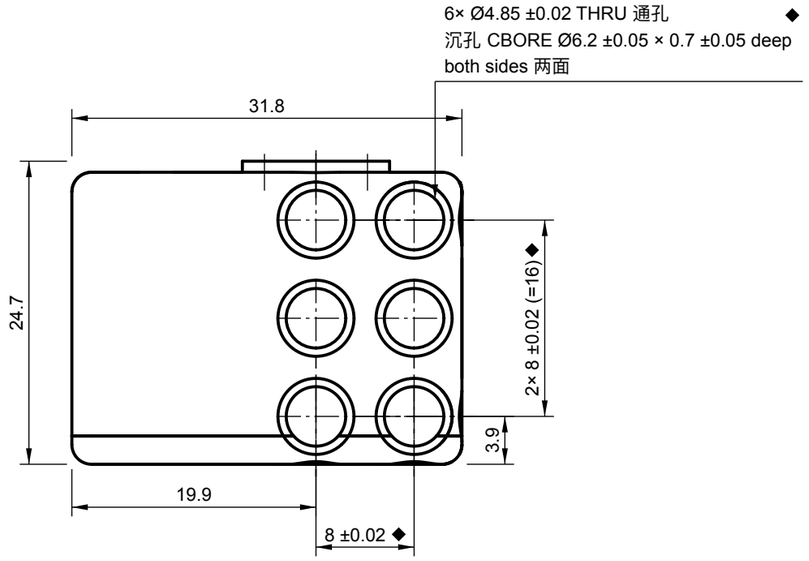
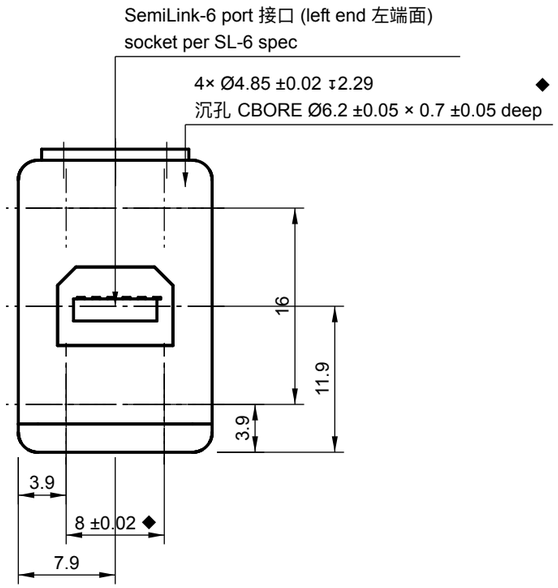
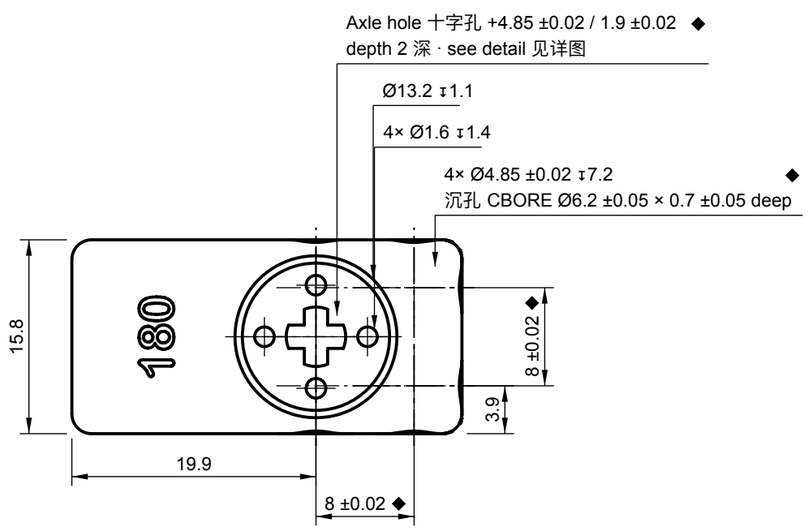
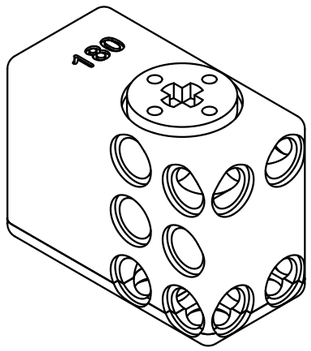






ISOMETRIC 轴测图 (1.5:1, ref.)



Technical requirements 技术要求

1. OUTLINE & INTERFACE drawing: fixes the envelope, the 8 mm pin-hole grid and the port position only. Enclosure internals (walls ≥ 1.2, PCB, bosses, snaps, battery) are designed by the supplier and approved by semiblock.ai.
外形及接口图：仅规定外形、8 mm 销孔网格和接口位置。外壳内部结构（壁厚≥1.2、PCB、螺柱、卡扣、电池）由供应商设计并经 semiblock.ai 批准。

2. Pin holes Ø4.85 ±0.02 ◆ on 8.00 ±0.02 ◆ pitch must accept SemiBlock and LEGO® Technic™ pins.
Ø4.85 销孔及 8 mm 孔距为关键尺寸，须兼容 SemiBlock 及乐高机械组销。

3. Electrical: SemiLink-6 port, ≤ 12 V. Safety: EN 62115 / GB 19865 (electric toys); radio modules CE RED / FCC / SRRC.
电气：SemiLink-6 接口，≤12 V。安全：EN 62115 / GB 19865 电玩具标准；无线模块需 CE RED / FCC / SRRC。

4. Material PC/ABS, UL94 V-0; colour Yellow 黄 (sRGB #FFC21A, std SB-C02); colour plaque approved before production.
材料：PC/ABS 阻燃 UL94 V-0；颜色 黄（#FFC21A，色号 SB-C02），量产前确认色板。

5. Material PC/ABS, UL94 V-0; colour White 白 (sRGB #F2F4F7, std SB-C03); colour plaque approved before production.
材料：PC/ABS 阻燃 UL94 V-0；颜色 白（#F2F4F7，色号 SB-C03），量产前确认色板。

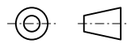
6. Pin holes: GO Ø4.83 / NO-GO Ø4.88 plug gauge. Fit test with SemiBlock and LEGO® Technic™ pins; push-in / pull-out force agreed at T1.
销孔用 Ø4.83 通规 / Ø4.88 止规检验；与 SemiBlock 及乐高销做装配测试，插拔力于 T1 确认。

7. Mark cavity number and material code >PC+ABS< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >PC+ABS<（ISO 11469），空间不足可省略。

8. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

9. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

10. Rev A is for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
A 版用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	PC/ABS UL94 V-0 PC/ABS 阻燃 V-0	**Weight** 重量	≈ 10 g housing
Colour 颜色	Yellow / White	**Qty / kit** 每套数量	1
Finish 表面	VDI 3400 Ref 18 细哑光纹理	**Scale** 比例	2:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	1 / 1

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
A	First issue for quotation & samples 首次发布 · 报价及打样	2026-10-27

Drawn 设计	semiblock.ai	2026-10-27
Checked 审核		
Approved 批准		

semiblock.ai SemiBlock Junior

Part No. 零件号

SB-SV180

Name 名称

SemiServo 180 deg (position) — outline 外形图

舵机 180°

Dwg SB-SV180 · Rev A · A3 · 3D: SB-SV180.step / .FCStd